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A Baseline method

To establish a benchmark for the challenge, we provide a multimodal baseline framework based on established deep learning architectures[1]. This framework aims to assist challenge participants in gaining a clear understanding of the expected performance standards and provide a reference point for evaluating their own models. As illustrated in Fig. 1, the system processes single-speaker video clips to simultaneously address Track 1 ("True" Personality Assessment) and Track 2 (Cognitive Ability Assessment). The pipeline of the system can be divided into three modalities.

- **Visual modality**: Captures non-verbal cues such as facial micro-expressions, gestures, and body language, which are pivotal for personality expression and reflecting cognitive effort or stress.
- **Audio modality**: Analyzes prosodic features, pitch, and vocal tone, which often correlate with emotional stability in personality and processing speed or fluency in cognitive tasks.
- **Verbal modality**: Analyzes the semantic content and linguistic structure of speech, reflecting both the lexical tradition of personality models (e.g., HEXACO) and the logical coherence indicative of cognitive ability.

The workflow consists of three stages: Feature Extraction, where modality-specific encoders compute local descriptors; Temporal Aggregation and Fusion, which compresses the temporal sequence into a unified multimodal vector; and Regression, where specialized heads predict the specific scores for both tracks.

A.1 Feature extraction

A.1.1 Visual modality. Visual behavior, particularly facial dynamics and posture, serves as a window into both an individual's trait-based tendencies and their real-time cognitive state. We utilize CLIP[2] (Contrastive Language-Image Pre-Training) to extract high-level semantic visual features. By leveraging its vision-transformer backbone, the model captures subtle cues in appearance and "visual affect" that are essential for predicting personality traits and detecting signs of cognitive load or engagement.

A.1.2 Audio modality. Acoustic characteristics provide vital clues regarding an individual's internal state. For Track 1, voice quality helps distinguish traits like *Extraversion* or *Emotionality*; for Track 2, it captures speech latency and phonation patterns related to cognitive processing. We employ whisper[3], a robust speech-processing model, to extract deep acoustic features. Unlike traditional handcrafted features, these embeddings capture a rich representation of the audio signal suited for diverse downstream regression tasks.

A.1.3 Verbal modality. The linguistic choices made by a speaker are perhaps the most direct indicators of both personality (through word choice and sentiment) and cognitive ability (through vocabulary diversity and syntactic complexity). We use RoBERTa[4] to process speech transcripts. This transformer-based model generates contextualized representations that capture the subtle nuances of dialogue, enabling the system to map language patterns to HEXACO personality dimensions and cognitive performance scores.

A.2 Modality fusion

To handle the asynchronous nature of the modalities, we apply temporal global average pooling to each stream. The resulting vectors are concatenated into a unified multimodal representation m in Fig. 1.

A.3 Regression

This joint embedding is then fed into the Regression Module. To ensure robustness and provide a measure of predictive uncertainty, we employ a Deep Ensemble approach. The module consists of 32 independent Multi-Layer Perceptrons (MLPs), each featuring hidden layers of 32 and 8 units. Separate regression heads are trained for Track 1 (predicting HEXACO traits) and Track 2 (predicting cognitive scores).

A.4 Results

Table 1 summarizes the performance of the baseline system across both tracks. For Track 1, we report the MSE for four key HEXACO personality traits. For Track 2, we provide the baseline accuracy for Cognitive Ability Assessment, establishing the difficulty level of the task.

[1]Code: https://github.com/AVIChallenge/AVI2026_baseline

[2]https://huggingface.co/openai/clip-vit-base-patch32
[3]https://github.com/openai/whisper
[4]https://huggingface.co/FacebookAI/roberta-base

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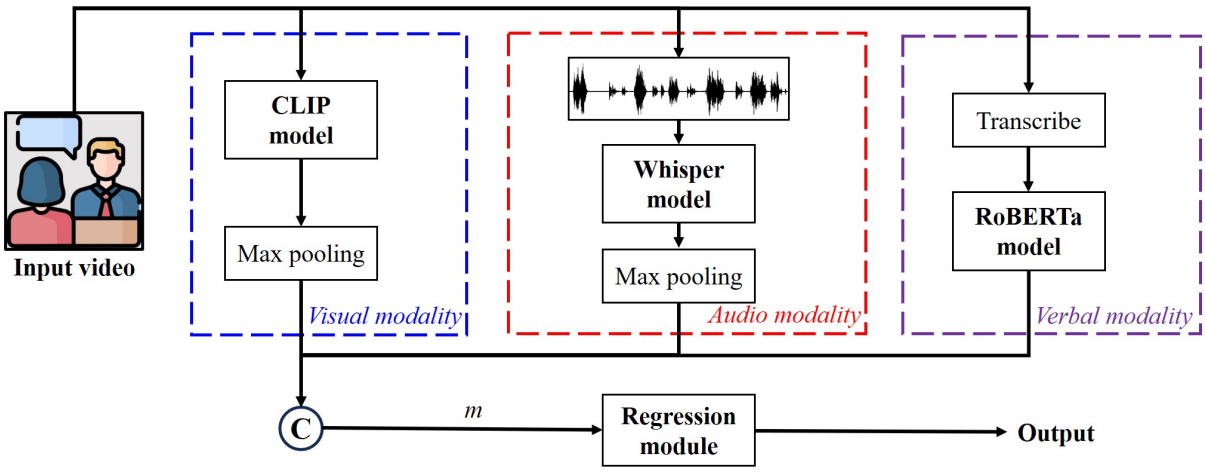

Figure 1. Pipeline overview: visual (blue), audio (red), verbal (purple) modalities, and 'C' stands for concatenation. During training, only the regression module requires personality annotations.

Table 1. Baseline performance for "True" Personality Assessment (*Track 1*) and Cognitive Ability Assessment (*Track 2*). Results for Track 1 are reported in Mean Squared Error (MSE ↓), and Track 2 in Accuracy (Acc ↑).

Method	**Track 1: "True" Personality (MSE)**					**Track 2: Cognitive Ability**
	H	E	A	C	*Avg.*	*Score (Acc)*
Proposed Baseline	0.2591	0.4347	0.3873	0.2524	0.3334	0.4062